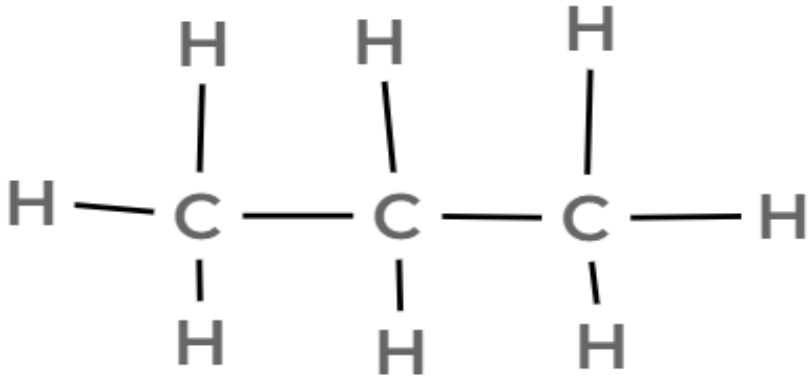
# CARD 01 — PROPANE

Molecular Formula: $C_3H_8$

## 1. MOLECULAR FORMULA

$C_3H_8$

## 2. LEWIS STRUCTURE




## 3. CONDENSED STRUCTURAL FORMULA

$CH_3CH_2CH_3$

## 4. BOND-LINE FORMULA (KETCHER)



☐ atom count verified ☐ valence checked ☐ bond order checked ☐ same connectivity confirmed

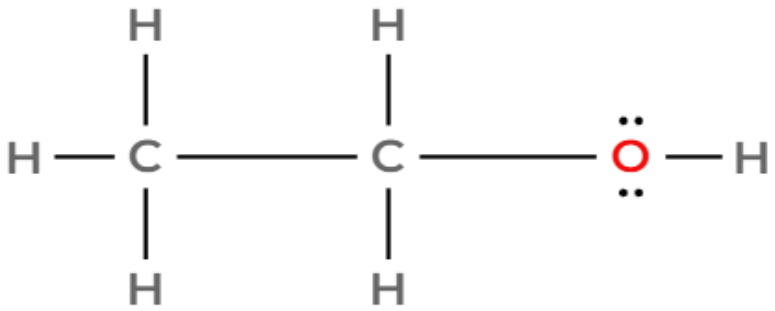
# CARD 02 — ETHANOL

Molecular Formula: $C_2H_6O$

### 1. MOLECULAR FORMULA

$$C_2H_6O$$

### 2. LEWIS STRUCTURE




### 3. CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2OH$$

### 4. BOND-LINE FORMULA (KETCHER)




☐ atom count verified ☐ valence checked ☐ bond order checked ☐ same connectivity confirmed

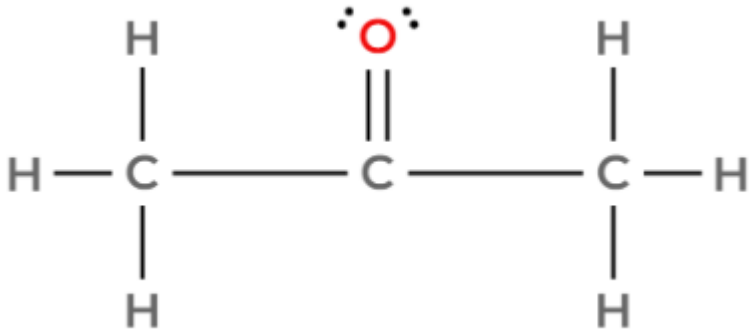
# CARD 03 — ACETONE (PROPAN-2-ONE)

Molecular Formula: $C_3H_6O$

### 1. MOLECULAR FORMULA

$$C_3H_6O$$

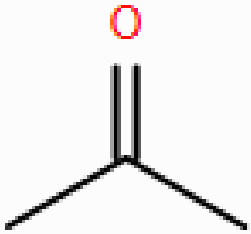
### 2. LEWIS STRUCTURE




### 3. CONDENSED STRUCTURAL FORMULA

$$CH_3COCH_3$$

### 4. BOND-LINE FORMULA (KETCHER)




☐ atom count verified ☐ valence checked ☐ bond order checked ☐ same connectivity confirmed

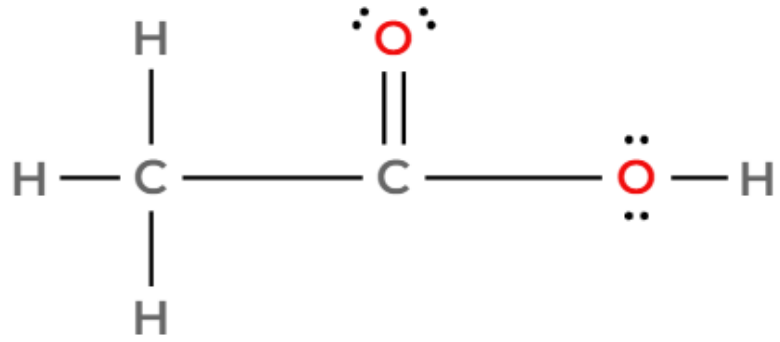
# CARD 04 — ACETIC ACID (ETHANOIC ACID)

Molecular Formula: $C_2H_4O_2$

### 1. MOLECULAR FORMULA

$$C_2H_4O_2$$

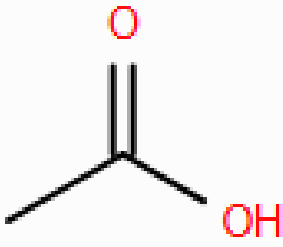
### 2. LEWIS STRUCTURE




### 3. CONDENSED STRUCTURAL FORMULA

$$CH_3COOH$$

### 4. BOND-LINE FORMULA (KETCHER)




☐ atom count verified ☐ valence checked ☐ bond order checked ☐ same connectivity confirmed

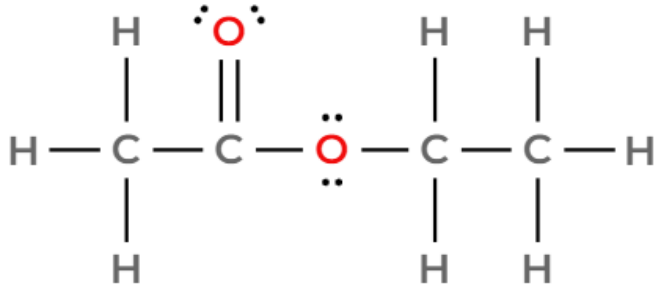
# CARD 05 — ETHYL ACETATE (ETHYL ETHANOATE)

Molecular Formula: $C_4H_8O_2$

### 1. MOLECULAR FORMULA

$$C_4H_8O_2$$

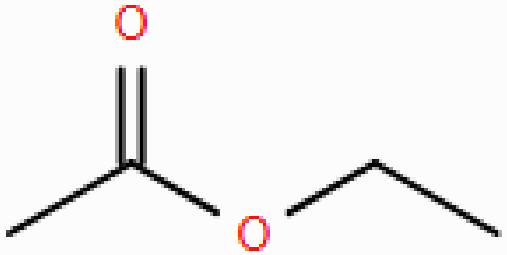
### 2. LEWIS STRUCTURE




### 3. CONDENSED STRUCTURAL FORMULA

$$CH_3COOCH_2CH_3$$

### 4. BOND-LINE FORMULA (KETCHER)




☐ atom count verified ☐ valence checked ☐ bond order checked ☐ same connectivity confirmed

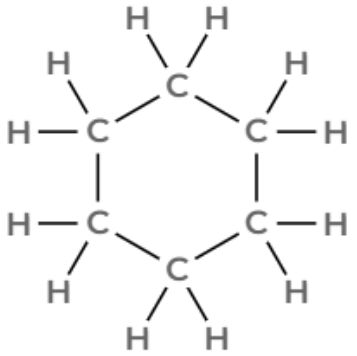
# CARD 06 — CYCLOHEXANE

Molecular Formula: $C_6H_{12}$

## 1. MOLECULAR FORMULA

$C_6H_{12}$

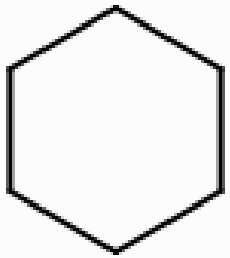
## 2. LEWIS STRUCTURE




## 3. CONDENSED STRUCTURAL FORMULA

$CH_2CH_2CH_2CH_2CH_2CH_2$

## 4. BOND-LINE FORMULA (KETCHER)



☐ atom count verified ☐ valence checked ☐ bond order checked ☐ same connectivity confirmed

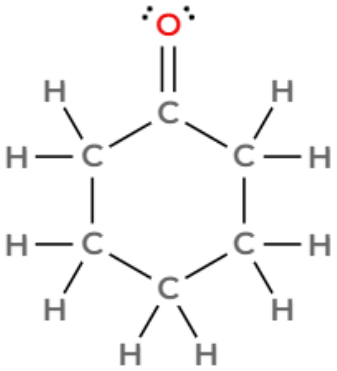
# CARD 07 — CYCLOHEXANONE

Molecular Formula: $C_6H_{10}O$

### 1. MOLECULAR FORMULA

$$C_6H_{10}O$$

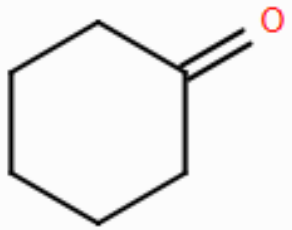
### 2. LEWIS STRUCTURE




### 3. CONDENSED STRUCTURAL FORMULA

$$(CH_2)_5C{=}O$$

### 4. BOND-LINE FORMULA (KETCHER)




☐ atom count verified ☐ valence checked ☐ bond order checked ☐ same connectivity confirmed

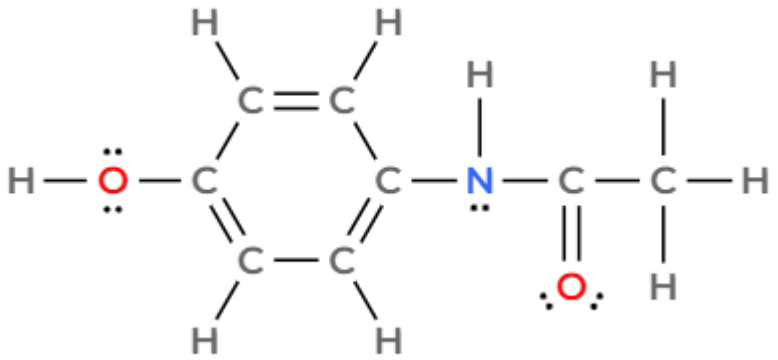
# CARD 08 — ACETAMINOPHEN (PARACETAMOL)

Molecular Formula: $C_8H_9NO_2$

### 1. MOLECULAR FORMULA

$$C_8H_9NO_2$$

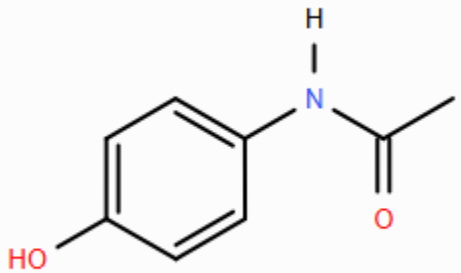
### 2. LEWIS STRUCTURE



### 3. CONDENSED STRUCTURAL FORMULA

$$CH_3CONH–C_6H_4–OH$$

### 4. BOND-LINE FORMULA (KETCHER)




☐ atom count verified ☐ valence checked ☐ bond order checked ☐ same connectivity confirmed

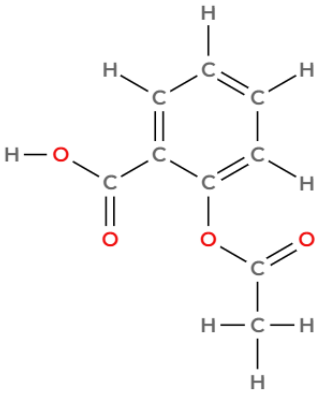
# CARD 09 — ASPIRIN (ACETYLSALICYLIC ACID)

Molecular Formula: $C_9H_8O_4$

## 1. MOLECULAR FORMULA

$$C_9H_8O_4$$

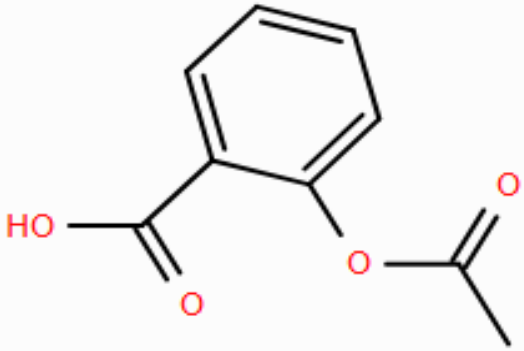
## 2. LEWIS STRUCTURE



## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3COO–C_6H_4–COOH$$

## 4. BOND-LINE FORMULA (KETCHER)




☐ atom count verified ☐ valence checked ☐ bond order checked ☐ same connectivity confirmed

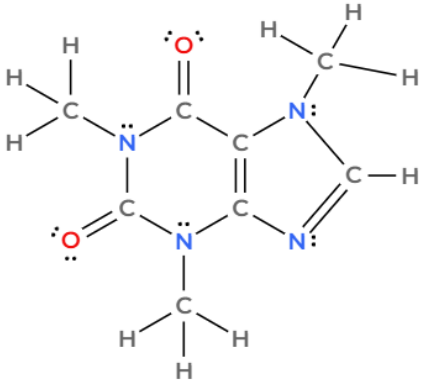
# CARD 10 — CAFFEINE

Molecular Formula: $C_8H_{10}N_4O_2$

## 1. MOLECULAR FORMULA

$$C_8H_{10}N_4O_2$$

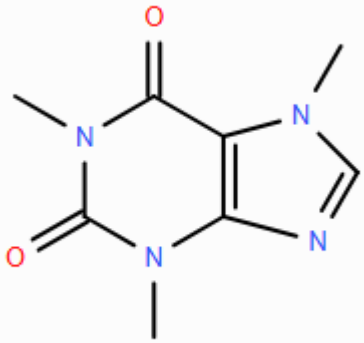
## 2. LEWIS STRUCTURE



## 3. CONDENSED STRUCTURAL FORMULA

$$CN_1C{=}NC_2{=}C_1C(=O)N(C(=O)N_2C)C$$

## 4. BOND-LINE FORMULA (KETCHER)



☐ atom count verified ☐ valence checked ☐ bond order checked ☐ same connectivity confirmed